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Effects of low-earth orbit exposure on geopolymer material properties

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Abstract

The construction of a lunar base camp including landing pads, radiation shielding, and transportation infrastructure is an integral part of NASA's upcoming Artemis missions. Geopolymer binders formed from locally available aluminosilicate-rich regolith on the moon and Mars are proposed as a robust solution for extraterrestrial construction materials. Lunar and Martian regolith simulants provide opportunities to synthesize candidate materials with optimized material properties. The goal of this work is to establish the durability of potential extraterrestrial geopolymer construction materials at low-Earth orbit conditions as an intermediate step towards the ultimate goal of testing on the lunar surface. As part of NASA's MISSE-20 mission, four compositions of geopolymer binders are exposed to low-earth orbit conditions (absolute temperature range of −11.75 °C to 35.0 °C) on the zenith position of the International Space Station for a duration of six months. Selected geopolymer compositions are synthesized from two lunar regolith simulants, one Martian regolith simulant, and high purity metakaolin. After samples return to Earth, the internal structure, compressive mechanical properties, and chemical composition are characterized and compared to control samples which remained on Earth for the duration of flight. Geopolymers formed from lunar regolith simulants Lunar Highlands 1 and Black Point 1 and the Martian regolith simulant MGS-1C show no signs of degradation resulting from low earth orbit exposure (LHS-1 sample compressive strength of 60.3 MPa vs Earth-control 44.7 MPa, ~35% increase). Samples formed from high purity metakaolin display visual darkening and cracking of the exposed material, determined to be a consequence of oxide attack and vacuum treatment during pre-flight testing, respectively. The results of this study explicitly demonstrate the viability of geopolymers as extraterrestrial construction materials and motivate the further development of methods to cure lunar geopolymers in environmental conditions relevant to the lunar surface.

Keywords: Geopolymer; In-situ resource utilization; Lunar construction; Low-earth orbit; Durability

1. Introduction

Durable extraterrestrial construction materials are required to build future lunar and Martian habitats, radiation shielding, landing pads, and transportation infrastructure along with other architectures as part of NASA's upcoming Artemis missions (Creech et al., 2022). To reduce payload costs from Earth, in-situ resource utilization (ISRU) or the use of locally available materials to create building materials is required (Anand et al., 2012; Sanders and Larson, 2012). Proposed technologies for the ISRU of lunar regolith for construction materials include geopolymer concrete (Alexiadis et al., 2017), sintered regolith (Farries et al., 2021), casted regolith (Naser, 2019), sulfur concrete (Gracia and Casanova, 1998), and regolith slag produced via molten regolith electrolysis (Schwandt et al., 2012). Geopolymer binders are the focus of this work, building on previous studies investigating the effects of both environmental conditions and varying chemical

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composition on the chemical structure and mechanical properties of lunar regolith simulant geopolymers (Egnaczyk et al., 2024; Mills et al., 2022).

Geopolymers are a class of alternative binder materials synthesized through the reaction of an aluminosilicate precursor and an alkaline solution (Duxson et al., 2007; Davidovits, 2017). Geopolymers formed from the aluminosilicate regolith provide a potential construction material requiring very little energy input to produce relative to alternative technologies as no components of the material need to be thermally melted (Spedding et al., 2020). The abundant lunar regolith is rich in aluminosilicate minerals providing ample reagent material without the need for purification (Papike et al., 1982). Importantly, the water required for geopolymer synthesis is not chemically integrated into the binder's final mineral-polymer structure and can potentially be recovered post-cure (Park and Pour-Ghaz, 2018; Wang et al., 2017). Conversely, cementitious materials like Ordinary Portland Cements form a hydrated reaction product which permanently sequesters water in the mix design. Many recent studies describe the synthesis of high strength geopolymers from lunar and Martian regolith simulant materials, supporting their potential for ISRU construction applications (Xiong et al., 2022; Mills et al., 2022; Zhou et al., 2021; Collins et al., 2022b).

Durability testing on the lunar surface is required to prove the aptitude of geopolymers for lunar construction applications and to advance the technology to a technology readiness level (TRL) of 5 (Mankins, 1995; ECSS, 2017). According to the current European Cooperation for Space Standardization (ECSS) standard, a TRL of 5 indicates that technology has been validated in a relevant environment, or the lunar surface in the case of lunar regolith geopolymers. In this work, the durability of geopolymer binders is investigated under accessible low-Earth orbit (LEO) conditions as an intermediate step towards testing on the lunar surface. Lab-based durability measurements are complicated by both the range of extreme conditions on the lunar surface and by the expansive geopolymer composition space available for study. Geopolymers on the lunar south pole will experience exposure to ultraviolet (UV) and gamma radiation, high energy charged particles like electrons and protons, pressures close to vacuum, and temperature swings from 50°K (−223 °C) to 202°K (−71 °C) at the lunar south pole (Williams et al., 2017). The cross-linked Si-O-Al bonds that form the geopolymer amorphous binder structure are highly resistant to gamma radiation-induced degradation when compared to the hydration products of Ordinary Portland Cement, which can develop porosity and micro-cracking under prolonged radiation exposure (Pomaro, 2016; Rosseel et al., 2016; Mast et al., 2019). Geopolymers derived from high purity metakaolin, a commonly used model geopolymer material formed from calcined kaolinite clay, darken when exposed to continuous UV and high energy electron radiation, yet display resistance to damage from intense gamma radiation (Cesul et al., 2014; Geddes et al., 2024). Geopolymers subjected to freeze–thaw cycle testing show a composition-dependent response, ranging from performance significantly worse than ordinary Portland cement (OPC) to significant improvements (Zhao et al., 2019; Fu et al., 2011; Jacobsen and Sellevold, 1996). Mills et al. observed high temperature exposure (600 °C for one hour) of lunar regolith simulant geopolymers, Lunar Highlands Simulant (LHS-1) and Black Point 1 (BP-1), increased the measured compressive strength. Inversely, Martian simulant geopolymers synthesized from clay-enriched Mars Global Simulant (MGS-1C) exhibited a decrease in strength likely attributable to the high iron content (Mills et al., 2022).

The strength development of geopolymers cured under vacuum conditions is significantly hindered, and methods for the initial curing of geopolymer binders in conditions relevant to LEO and the lunar surface are an active area of investigation (Davis et al., 2017; Collins et al., 2022a). The compressive strength of volcanic ash based geopolymers is observed to decrease following exposure to vacuum conditions after an initial precure period (Wang et al., 2017). Alternatively, the compressive strength of metakaolin geopolymers is reportedly unaffected by combined vacuum and elevated temperature for a period of 24 h, while the BET surface area of vacuum-treated samples increased (Li et al., 2017a). Like many measurements of geopolymer material properties, the impact of vacuum exposure on compressive strength is highly dependent on composition and processing or curing methods.

The Multi-purpose ISS Experiment Flight Facility (MISSE-FF) missions provide an opportunity to test the performance and durability of materials and devices exposed to the low Earth orbit space environment (De Groh and Banks, 2021). As part of the MISSE-20 mission, this study represents the first exposure of geopolymer binders on the ISS exterior for the purpose of degradation testing, as an intermediate step towards testing on the lunar surface. The goal of this work is to establish the durability of potential extraterrestrial geopolymer construction materials following exposure to low-Earth orbit conditions for a period of six months. Through a combination of imaging, mechanical testing, and assessment of chemical composition, the impact of extended LEO exposure on chemical and mechanical properties of representative geopolymer binders is assessed and compared to control groups which remained on Earth. The effects of geopolymer composition on durability are studied via the measurement of geopolymers formed from the lunar regolith simulants LHS-1 and BP-1, the Martian Simulant MGS-1C, and a standard metakaolin geopolymer.

2. Materials and methods

2.1. Geopolymer compositions

Four geopolymer compositions are prepared from aluminosilicate precursor materials including the lunar regolith simulant Lunar Highlands Simulant 1 (LHS-1, Space Resource Technologies), the Black Point 1 lunar mare simulant (BP-1, NASA Kennedy Space Center Swamp Works Group), the Martian regolith simulant Mars Global Hydrated Clay Simulant (MGS-1C, Space Resource Technologies) and a commercial metakaolin (MasterLife MK828 Metakaolin (MK)).

The LHS-1 simulant mimics the chemical composition and particle size distribution of lunar highlands soils brought back by the Apollo 16 mission (Fig. 1A) (Long-Fox et al., 2023). Important for Artemis applications, the highlands type regolith is found at the lunar south pole which has been identified as a target for the Artemis base camp. BP-1 is a volcanic basalt-derived aluminosilicate powder sourced from the Black Point basalt flow in the San Francisco Volcanic Field in northern Arizona (Fig. 1B). BP-1 is used as a geotechnical simulant by NASA for annual Lunabotics compositions, as the particle size distribution of BP-1 closely resembles Apollo lunar regolith samples (Stoeser et al., 2010). The MGS-1C simulant is a clay-enriched variant, based on analysis from Curiosity rover's onboard instruments including X-ray diffraction (Fig. 1C) (Long-Fox and Britt, 2023). The metakaolin based geopolymer is included as a control composition due to the high aluminosilicate purity (Fig. 1D) (>95% SiO_2 and Al_2O_3). The compositions of each aluminosilicate precursor are listed in Table 1. BP-1, LHS-1, and MGS-1C simulants are sieved with a #200 (75 μm) mesh prior to geopolymer synthesis, and the average particle size (d_{50}) of the metakaolin is ~5 μm.

Compositional differences in the aluminosilicate powders emphasize the range of potential chemistries accessible for extraterrestrial ISRU-based construction materials. MGS-1C is notably rich in iron and magnesium oxides. LHS-1 is relatively poorer in magnesium and iron compared to the BP-1 mare regolith simulant. A wide range of aluminum content is present in the selected aluminosilicates, ranging from 40 wt% in the metakaolin to only 9.8 wt% in the MGS-1C. Similarly, CaO content varies widely from almost zero in the metakaolin to >13 wt% in the LHS-1 and MGS-1C aluminosilicates.

Fig. 1. Aluminosilicate powders used for geopolymer synthesis including (A) Lunar Highlands Simulant 1 (LHS-1), (B) Black Point 1 (BP-1), (C) Mars Global Hydrated Clay Simulant (MGS-1C), and (D) metakaolin.

Table 1
Composition of aluminosilicate precursor materials.

Oxide	LHS-1 (Exolith)	BP-1 (Stoeser)	MGS-1C (Exolith)	MK
SiO_2	49.12	47.2	44.8	55
TiO_2	0.63	2.3	0.4	1.5
Al_2O_3	26.29	16.7	9.8	40
FeO	3.2	6.2		
MgO	2.86	6.5	9.9	
CaO	13.52	9.2	15.3	0.1
Na_2O	2.55	3.5		0.2
K_2O	0.34	1.1	3.4	0.62
P_2O_5	0.17	0.52	1	
Fe_2O_3		5.9	12	1.4
SO_3			2.3	
Cl			0.5	
Total	99.15	99.9	98.11	98.82

Sodium silicate activating solutions are prepared from deionized water (Millipore Milli-Q 18.2 MΩ), sodium hydroxide pellets (Fisher Scientific CAS:1310-73-2, 497-19-8), and fumed silica (Cabot CAB-O-SIL EH-5 Hydrophilic Fumed Silica). Sodium hydroxide pellets are mixed into the water using a Teflon-coated magnetic stir bar for thirty minutes while sealed in a stainless-steel beaker. Fumed silica is slowly incorporated into the solution and resealed. The activating solutions equilibrate for a minimum of 12 h before geopolymer synthesis at room temperature (22 ± 2 °C) while constant stirring of the solution is maintained via the Teflon stir bar at a stirring speed of 150 rotations per minute. Two different activating solution compositions are used in this study (Table 2). Composition I is used for the formation of BP-1, LHS-1, and MGS-1C geopolymers and Composition II is used for the metakaolin geopolymer (Egnaczyk et al., 2025).

Geopolymer compositions are selected to develop a high compressive strength, and the per-gram formulations are listed in Table 3. LHS-1 and BP-1 geopolymers are composed of 70 wt% aluminosilicate and 30 wt% activating solution composition I and the MGS-1C geopolymer is

Table 2
Sodium silicate activating solution compositions on a per-gram basis.

Composition	Mass H_2O	Mass NaOH	Mass SiO2	Si/(Si + Na)
I	0.6	0.16	0.24	0.5
II	0.52	0.21	0.27	0.5

Table 3
Geopolymer compositions on a per-gram basis.

Geopolymer	Mass Aluminosilicate	Mass H_2O	Mass NaOH	Mass SiO_2
LHS-1	0.7	0.18	0.05	0.07
BP-1	0.7	0.18	0.05	0.07
MGS-1C	0.6	0.24	0.06	0.10
MK	0.4	0.31	0.13	0.16

Table 4
Bulk densities of geopolymer samples. The Immediate Control, the Earth Control samples, and the post-flight MISSE samples are assessed for a minimum of four samples each.

Aluminosilicate source	Immediate control density [g/cm^3]	Earth-control bulk density [g/cm^3]	MISSE sample bulk density [g/cm^3]
Black Point 1	2.03 ± 0.03	2.03 ± 0.02	2.04 ± 0.01
Lunar Highlands 1	1.74 ± 0.02	1.77 ± 0.01	1.78 ± 0.01
Mars Global Hydrated Clay Simulant	2.08 ± 0.04	2.11 ± 0.04	2.10 ± 0.03
Metakaolin	1.33 ± 0.01	1.36 ± 0.02	1.35 ± 0.01 *

* Only two sample densities are measured due to severe cracking.

composed of 60 wt% MGS-1C and 40 wt% activating solution composition I. The MGS-1C formulation was chosen based on previous results from Mills et al., which indicates the need for higher activating solution content to form a consolidated binder (Mills et al., 2022). The MK geopolymer is formulated with activating solution composition II containing 40 wt% MK and 60 wt% activating solution to achieve a desired Si/Al/Na/H2O ratio of 2/1/1/5.5.

2.2. Curing protocol

A uniform mixing protocol is used for all geopolymer samples. After one minute of hand mixing to incorporate the liquid and powder components, samples are mixed for four minutes at a rotation speed of 500 rpm using an IKA RW 20 digital mixer. The geopolymer pastes are poured into five Teflon molds each containing four rectangular molds of dimensions 2.54 × 2.54 × 1.02 cm (1 × 1 × 0.4 in.). Teflon molds are custom made for this study to achieve set dimensions for MISSE-20 integration, with sample mold corners to a radius of 0.1 cm. The screw assembly of the molds achieves a water-tight seal. Sets of 20 uniform geopolymer slabs are synthesized from a single batch of activating solution at the same time to reduce experimental variation. The sample synthesis and cure protocol is shown in Schematic A in Fig. 2.

After mixing and molding, an identical cure protocol was applied to all samples. (1) After the mixed geopolymer paste is poured into the mold, the molds are sealed airtight. (2) Sealed molds are placed in an oven pre-heated to 60 °C and maintained at this temperature for a period of 72 h. (3) The molds are removed from the oven and the geopolymer slabs are removed from the molds and placed back into the pre-heated oven at 60 °C for an additional 72 h at constant temperature. (4) The geopolymer slabs are removed and sanded sequentially with 80, 120, and 180 grit sandpaper to the specifications required for the MISSE-20 experiment.

After the synthesis and curing procedure is complete for each composition sample set, samples are rearranged into five final groups to eliminate any potential mold-specific effects from influencing final experimental results, shown in Schematic B of Fig. 2. The following protocol is followed

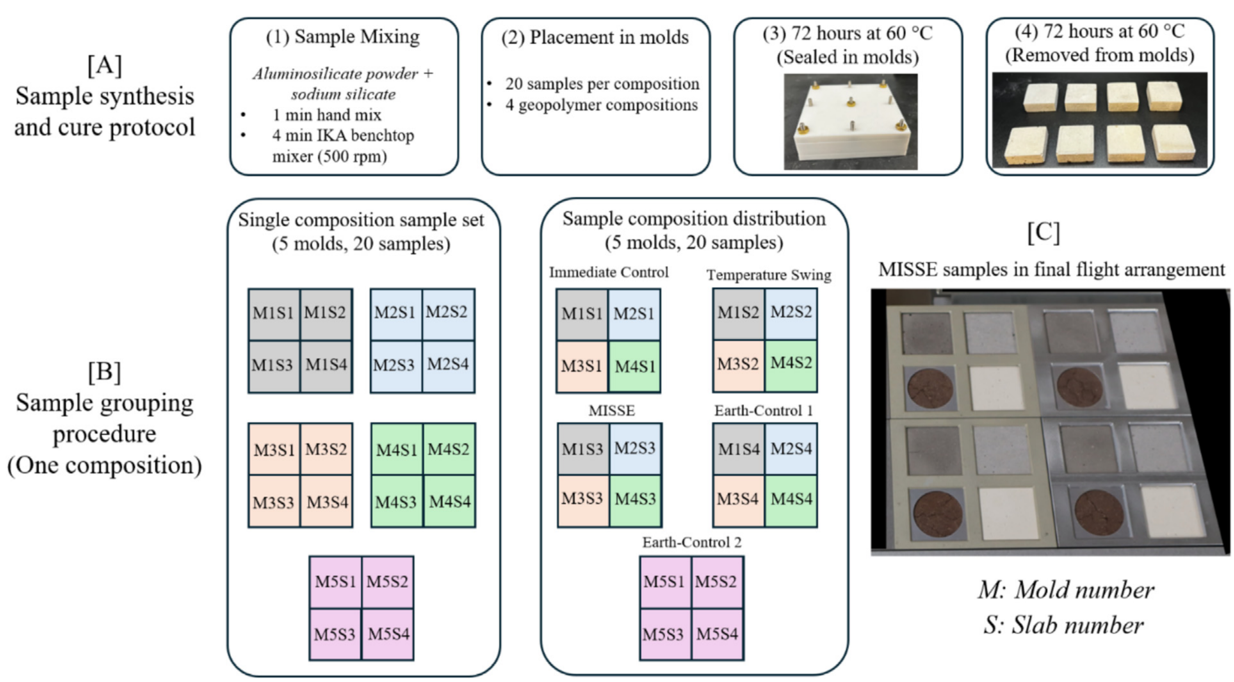

Fig. 2. Schematic of sample preparation procedure and grouping assignments. (A) Sample synthesis and cure protocol workflow. (B) Sample grouping procedure. The example grouping diagram details the arrangement of the 20 samples created for a single geopolymer composition. (C) Image of final MISSE sample arrangement in the sample holder showing four samples included for each of the four geopolymer compositions.

for each 20-slab set, representing a single geopolymer composition. After sanding, the geopolymers from the five molds are reorganized into five groups. One slab from each of the first four molds is combined to form four new sets of four slabs to eliminate any mold-specific effects. Samples from the fifth mold are retained separately. For each composition, five final sets of four slabs were created including the MISSE-20 experimental sample set ("MISSE samples", four samples), two Earth Control sample sets ("Earth-control samples", eight samples per-composition), a set for immediate compressive strength (CS) testing ("Immediate control samples", four samples), and a set for CS testing after exposure to simulated "bake-out" testing ("Temperature-swing samples", four samples). The final array of samples prepared for the MISSE-FF mission contains four samples of each of the four compositions composition (Fig. 2C). Prior to any sample analysis and following sanding, the dimensions and masses of all geopolymer samples are recorded using a Neiko digital caliper (dimensional tolerance of final samples $\leq \pm 0.02$ mm) to calculate bulk sample density and test for shrinkage during the curing process. *For the remainder of this manuscript, "Immediate Control" samples refer to samples which undergo material property testing seven days after synthesis without any additional treatment. "Temperature-swing samples" refers to the samples which underwent controlled bake-out testing at the University of Delaware seven days after synthesis. "MISSE samples" refers to geopolymer samples which were exposed to LEO conditions as part of the MISSE-20 mission. "Earth-control samples" refers to the sample sets synthesized in tandem with the MISSE samples and stored in blackout bags for the duration of the MISSE-20 mission.*

2.3. Bake-out testing

MISSE samples are subjected to "bakeout testing" prior to flight, a two-part sequence of tests consisting of thermal and pressure swings meant to ensure that materials will withstand the conditions of flight to the ISS along with minimal de-gassing under vacuum conditions. Step one of the pre-flight bakeout protocol performed by Aegis Aerospace included 24 h of equilibration at 1×10^{-5} Torr at 60 °C. Next, a thermal ramp protocol was performed, consisting of a temperature ramp from 55 to −46 °C with a ramp rate of 3 °C/min and a dwell time of one hour at the temperature extremes. The thermal ramp protocol is initiated and ended at ambient temperature and ambient pressure is maintained throughout.

The temperature-swing geopolymer sample sets undergo an analogous testing sequence at the University of Delaware to assess the influence of this conditioning on their mechanical performance. Seven days after synthesis, the specimens are placed in a vacuum oven at 60 °C for 24 h. Immediately after removal, the specimens are subjected to a controlled temperature-cycling regimen at atmospheric pressure consisting of: (1) one hour at standard laboratory temperature (20 ± 1 °C), (2) one hour in a dry-ice chamber for low-temperature exposure (–78 °C), (3) one hour returned to laboratory temperature, and (4) one hour in a convection oven at 60 °C.

2.4. MISSE-20 mission description

In total, the MISSE samples experienced 201 days of exposure in space, traveling over 82 million miles and completing 3216 orbits of earth in the Zenith 2 mount orientation aboard the International Space Station. The entire experiment represents a two-year effort. Following a call for proposals in July of 2023, the project proposal was accepted in October of 2023. Geopolymer samples were created in the months of March-April 2024 and shipped to Aegis Aerospace in Webster, TX for integration into the MISSE-20 sample carriers. Following bake-out testing performed by Aegis Aerospace, samples were flown to the ISS as part of the Space X CRS-31 commercial resupply service flight on November 5, 2024. From November 2024 – May 2025 the samples were exposed on the ISS. Finally, during May of 2025 the samples returned to Earth on SPX-32 and were returned to the University of Delaware for final testing in July 2025.

During the flight MISSE samples were held in place by modular sample cases. LHS-1, BP-1, and metakaolin geopolymer samples were held in cases with square openings of 2.15 × 2.15 cm. MGS-1C geopolymer samples are mounted in cases with 2.15 cm diameter circular exposure openings due to shrinkage of the material during the cure process. Images of select samples were collected for the months of November 2024 through February 2025. Additional temperature data was collected continuously, and the UV radiation on the sample was also measured for the entire flight duration. The value of total UV energy absorbed by each cube is calculated by integration of the provided UV exposure data over time.

2.5. X-ray tomography

X-ray tomography scans are collected for all MISSE samples and select Earth-control samples for comparison of the internal structural features. A Rigaku GX 130 CT Scanner is used to collect three-dimensional reconstructions of the sample in a series of X-ray micro-CT scans. A field of view of 45 mm captures the entire geopolymer sample volume in a single scan. Instrument parameters are set to a voltage of 130 kV, a current of 300 μA, and a scan time of 14 min in high resolution mode. Representative sample images and videos are created in the Dragonfly 3D World software.

2.6. Compression testing

Compressive strength measurements for the Immediate Control and Temperature Swing samples are performed on an Instron model 1331 servo hydraulic test frame with a 100kN Instron 2518-611 load cell. Compressive strength measurements for the MISSE and Earth Control samples are performed using an Instron model 5985 system R9874 with an Instron 250 kN load cell. An identical protocol is followed for both measurement sets. Geopolymer slabs are compressed between two stainless steel platens using an augmented ASTM C109 protocol, Compression testing of hydraulic cement mortars (ASTM International, 2020). The testing protocol differs from the ASTM standard in the size of the sample material (2″ × 2″ × 0.4″ slabs) and the cure time determined by the MISSE-20 mission schedule. The compression rate is 1 mm/min. The measured force–displacement curves are transformed to engineering stress–strain curves according to Eqs. (1) and (2), where σ is the engineering stress, F is the measured force, A is the sample surface area, ε is the engineering strain, Δh is the displacement, and h is the height of the geopolymer slab. The compliance of the measurement system in the absence of a sample is measured in a separate test following sample testing. A force vs displacement curve is collected at a compression rate of 0.2 mm/min is collected for the stainless-steel platens without any sample present up to a maximum applied force of 80 kN, exceeding the maximum force required for sample compression. A cubic equation is then fit to the displacement vs force data (R^2: 0.999) to determine the instrument compliance as a function of applied force. The fitted instrument compliance as a function of applied force is then subtracted from the machine displacement to determine the true sample displacement at a given applied force.

$$\sigma = F/A \tag{1}$$

$$\varepsilon = \Delta h/h \tag{2}$$

The maximum compressive strength is determined for each individual geopolymer and averaged for each composition and sample exposure condition. Compressive strength is determined via one of two methods depending on the stress–strain behavior of the samples. The compressive strength is selected as the maximum compressive stress for samples which display a clear fracture behavior with a defined stress peak. A subset of geopolymer compositions displayed a segmented response where the fracture of the material is followed by a crumbling-like response leading to two slopes in the data. For the second sample type, a linear fit is applied to each stress–strain regime, and the compressive strength is assessed as the engineering stress at the intersection point of these two lines. An example plot depicting the two methods for determining CS is available in the Supplementary Material.

The strain to-failure is determined as the corresponding strain value to the maximum compressive strength. The secant modulus, a measure of the elasticity or Young's modulus, was determined according to ASTM C469, Static Modulus of Elasticity and Poisson's Ratio of Concrete in Compression (ASTM International, 2022), with Eq. (3) below, where E is the secant modulus, ε_1 is a strain of 0.00005, σ_1 is the corresponding stress to ε_1, σ_2 is 40% of the maximum compressive strength, and ε_2 is the corresponding strain at the maximum compressive strength.

$$E = \frac{(\sigma_2 - \sigma_1)}{(\varepsilon_2 - \varepsilon_1)} \tag{3}$$

2.7. X-ray diffraction measurements

XRD patterns are collected for the LHS-1 aluminosilicate precursor, MISSE samples, and Earth-control samples. Patterns are also collected for the metakaolin aluminosilicate precursor, MISSE samples, and Earth-control samples. X-ray diffraction (XRD) measurements are performed on a Bruker D8 Discover powder diffractometer equipped with a 1.54 Å Cu Kα radiation source and a 1.2-mm slit. A step time of 1.5 s per step and a step size of 0.05° are employed, and the scan sequence is repeated three times for each specimen to ensure an adequate signal-to-noise ratio. Diffraction patterns are acquired in 1D mode over a 2θ range of 5–70°. The resulting diffraction profiles are processed using Bruker DIFFRAC.EVA software with minimal background correction.

3. Results

3.1. Effects of pre-flight bakeout testing

Metakaolin geopolymer samples cracked during bakeout testing, both for the MISSE samples subjected to preflight bake-out testing protocol performed by both Aegis Aerospace and temperature-swing samples at the University of Delaware. Testing at the University of Delaware is performed while the samples are not loaded into sample holders, providing clear visual evidence of large cracks spanning the entire samples (Fig. 3A). Pre-flight bakeout testing at Aegis is performed while the samples are loaded into the sample holders (Fig. 3B), showing similar macroscale cracks for two of the four samples, and hairline cracks on the other two samples. The LHS-1, BP-1, and MGS-1C samples do not show any signs of degradation or cracking following the bakeout testing.

3.2. Surface morphology

Visual changes on the surface of some MISSE geopolymer samples are present following LEO exposure, while others are unchanged. LHS-1 and BP-1 geopolymer samples did not experience significant discoloration (Fig. 4 A-D), except for a thin rectangular outline left from the contact of the MISSE sample holder with the sample. Comparing the Earth control and MISSE MGS-1C samples (Fig. 4 E vs F) the circular portion of MISSE samples exposed to LEO is darker than the portion covered by the sample holder. The MISSE metakaolin geopolymers exposed to LEO conditions are a significantly darker shade (Fig. 4 G-H). A clear difference is visible on the surface of all metakaolin MISSE samples when comparing the lighter

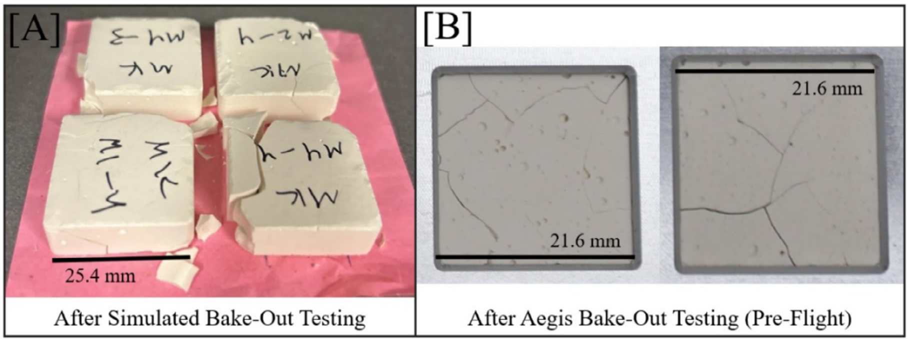

Fig. 3. Images of metakaolin geopolymer samples following (A) simulated bakeout testing at the University of Delaware and (B) pre-flight bakeout testing performed by Aegis Aerospace. Images from Egnaczyk and Wagner, J. Rheology, 2025 used with permission.

edges covered by the sample holder to the central exposed surface. Measurements of UV radiation during flight indicate that all MISE samples absorbed 15.0 kJ/cm^2 of UV energy, equivalent to 69.1 kJ per exposed sample. All MISSE samples experienced an absolute temperature range of −11.75 to 35.0 °C. During each of the 16 daily revolutions of Earth, sample temperature changed by ~15 °C. Compiled radiation and temperature data during flight are available in the Supplementary Material.

The post-flight surface topography of the geopolymer samples is also differentiated by the aluminosilicate type. LHS-1 and BP-1 MISSE samples do not display any visible cracking behavior. Macroscopic cracks transversing the entire sample are observed in the metakaolin MISSE samples only. Visible crevices are present throughout the MGS-1C geopolymers; however, these are present in both the Earth Control and MISSE samples and are likely due to shrinkage of the material during the elevated temperature curing process. The MGS-1C crevices are not sample-spanning, and the samples are a continuous solid.

In-flight images of the MISSE geopolymer samples (Fig. 5) support results from the post-flight visual inspection. BP-1 and LHS-1 MISSE samples do not show any visual signs of degradation or color change in the first three months of exposure. MISSE metakaolin (MK) geopolymers contain visible cracks in the earliest available ISS images, following the cracking event during bakeout testing. As the flight progresses, no additional cracks develop but darkening of the sample is clearly visible from month 1 to month 3 of exposure (Fig. 5E-F).

3.3. Internal structure

A difference in the internal morphology of geopolymer samples is only observed between MISSE and Earth Control samples for metakaolin geopolymers. No clear differences in the internal structure of the LHS-1, BP-1, or MGS-1C samples between the Earth Control and MISSE samples are evident (Fig. 6). Scans through the MGS-1C samples show that the visible crevices are not continuous throughout the samples, contrasting with the sample-spanning cracks in the metakaolin. Videos of x-ray tomography scans vertically through representative MISSE sample, and Earth control samples are available in the Supplementary Material.

All samples contain some residual air pockets regardless of composition, likely due to the high shear mixing of the initial samples. The high rheological yield stress of the initial sample pastes prevents air pockets from escaping despite repeated tamping of the materials while molded. Hairline edge cracks which do not span the entire sample are present in both the Earth Control and MISSE BP-1 geopolymer samples but are only observable in internal tomography scans.

3.4. Material properties

The bulk densities of geopolymer samples are equivalent within one standard deviation when measured at the three conditions of Immediate Control, the Earth-control samples, and the post-flight MISSE samples. Water originally present in the samples likely evaporates during the second stage of the curing process except for any bound water in the crosslinked geopolymer molecular network or trapped in the nanopores of the geopolymer matrix.

Compression testing of the geopolymer samples provides a measurement of the sample compressive strength, strain to failure, and secant modulus. Compressive strength results are presented in Fig. 7 A-D, and the corresponding strain to failure and secant modulus data is provided in the Supplementary Material. Compiled engineering stress–strain curves for each composition are also available in the Supplementary Material. All immediate control samples formed high-strength materials with compressive strengths exceeding 35 MPa.

First, a statistical comparison of the material property results of samples before and after both types of bakeout testing is performed via a series of two-tailed t-tests using two sample unequal variance (Table 5). The compressive strength of LHS-1 geopolymers significantly increases following both the bakeout testing (Temp Swing vs Immediate Control in Fig. 7A) and following MISSE flight (MISSE vs Earth Control in Fig. 7A). An alpha level of 0.05 is used for all t-tests. Results in Table 5 indicate that

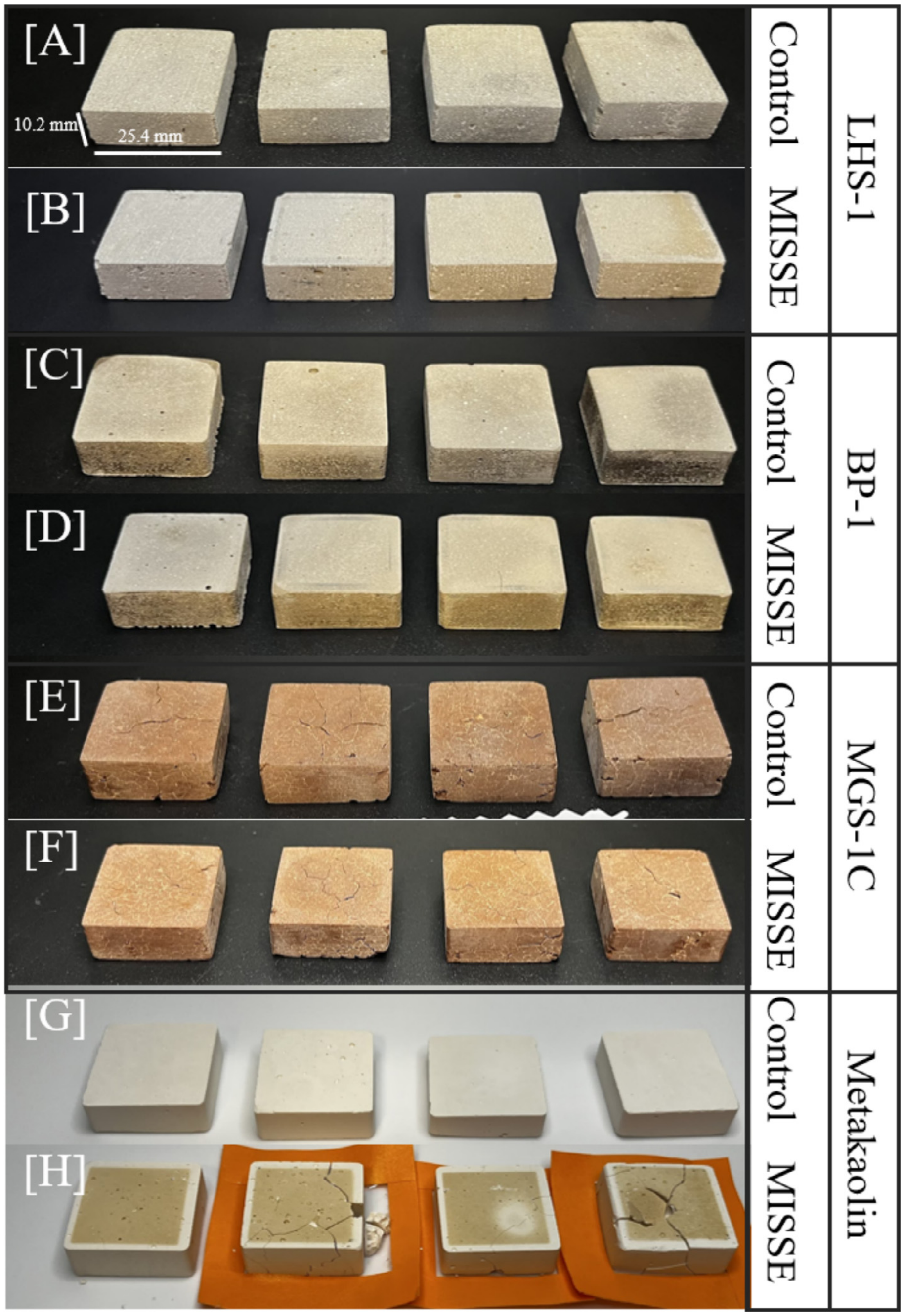

Fig. 4. Visual comparison of MISSE samples and earth-control samples following MISSE sample return. Sets of four samples are shown including (A) LHS-1 control, (B) LHS-1 MISSE, (C) BP-1 control, (D) BP-1 MISSE, (E) MGS-1C control, (F) MGS-1C MISSE, (G) Metakaolin control, and (H) metakaolin MISSE samples. Metakaolin MISSE samples which experienced sample-spanning fracture are placed on orange tape to prevent separation of the segments. All photographs are captured using a white LED overhead lamp and no editing or visual effects were applied.

the measured compressive strength of the MISSE LHS-1 geopolymers ($M = 60.3$, $SD = 2.8$) is higher than the Earth Control LHS-1 samples ($M = 44.7$, $SD = 2.0$), $p = 0.00027$. Significantly higher compressive strength is also measured for BP-1 MISSE samples compared to the Earth Control samples. Comparing the Immediate Control samples to the Earth Control samples, a significant increase is observed for the LHS-1, MGS-1C, and Metakaolin geopolymers.

Images of the Earth-control and MISSE samples before and after compression in Figs. 8 and 9 show different crushing failure mechanisms for each aluminosilicate type, substantiated by the trends present in the compiled engineering stress–strain curves. Earth-control and MISSE LHS-1 samples all displayed axial splitting with larger sample chunks splitting from the bulk, and a corresponding clear maximum for all samples in the stress–strain curves. Earth-control and MISSE BP-1 samples compress in a

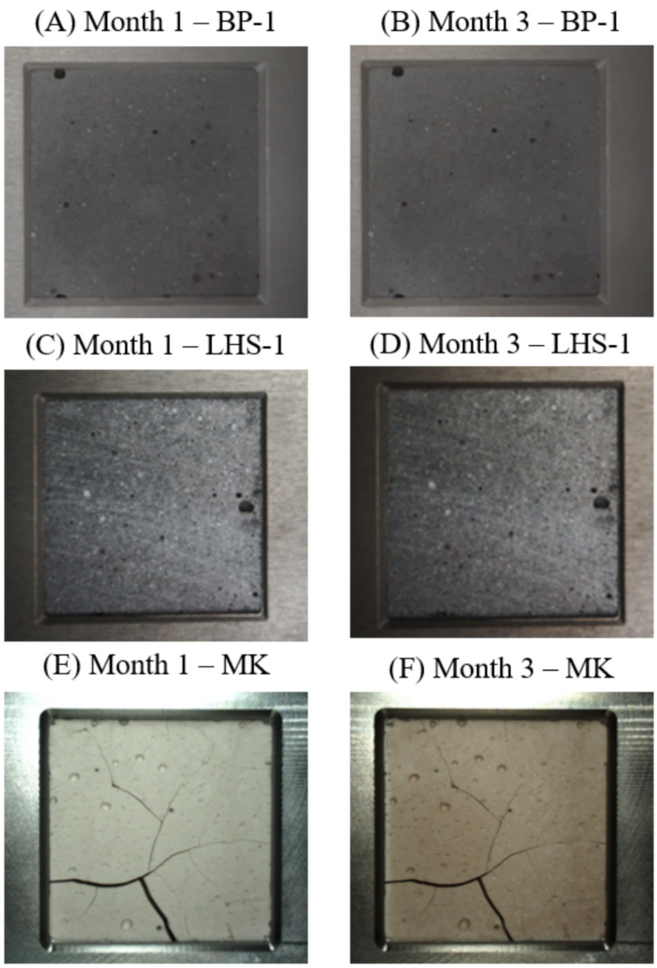

Fig. 5. In-flight images of geopolymer sample surface exposed to LEO conditions. Representative samples include a BP-1 geopolymer sample following (A) one month of exposure and (B) three months of exposure, a LHS-1 sample following (C) one month of exposure and (D) three months of exposure, and a metakaolin sample following (E) one month of exposure and (F) three months of exposure. Images uniformly edited for brightness.

two-step process where failure after initial elastic compression is followed by a slow crumbling-like response. The BP-1 stress–strain curves contain two different regimes indicative of a compression followed by the crumbling response after the major failure of the material. Visually, the Earth-control and MISSE MGS-1C samples also show a crumbling-like response to compression, but the stress–strain curve data shows a clear peak for the majority of samples. However, the crevice-filled nature of the samples leads to less-consistent sample loading and more noise-filled stress–strain data leading to increased variation in the calculated material properties. Last, the metakaolin Earth Control samples failed in a single mode, creating clear axial splitting leading to shard-like remnants (Fig. 9D). The sample-spanning fractures present in the MISSE metakaolin sample set also lead to the relatively large standard deviation in compressive strength. Differences in the compression mechanism of the MISSE and Earth Control samples is only apparent for the metakaolin samples (Figs. 8D and 9D). The MISSE metakaolin samples crumble rather than splitting axially, creating small almost cubic pieces of debris, likely due to the presence of many micro cracks in the sample.

3.5. Chemical structure

XRD analysis of the aluminosilicate precursor, MISSE sample, and Earth-control sample is performed for the metakaolin and LHS-1 geopolymers to compare chemical changes in the binder with the significant changes in material properties. Relative comparison of the scans is prioritized to highlight the effects of the different sample treatments, rather than identification of the crystalline species present in each scan. XRD spectra of the MISSE metakaolin geopolymer and Earth Control metakaolin geopolymer show identical features (Fig. 10A). In comparison to the metakaolin aluminosilicate powder, both geopolymer spectra show a shift in the amorphous hump center from ~22 to ~28 degrees (2θ) indicating the formation of amorphous polymerized reaction product (Provis et al., 2005). A peak at 25.3 degrees (2θ) indicative of quartz is present in all samples. A statistical comparison of the XRD patterns is achieved through a Root Mean Square Error (RMSE) analysis comparing pairs of XRD patterns, shown in Table 6. The RMSE value of 0.37 for the comparison of the MISSE and Earth Control metakaolin samples indicates a very close match of XRD pattern features compared to the relatively large value of 2.9 resulting from the comparison of patterns from the Earth Control and aluminosilicate metakaolin precursor. The similarity of Earth Control and MISSE XRD patterns indicates that few chemical changes occurred in the binder during LEO exposure.

Chemical analysis of the MISSE and Earth-control LHS-1 geopolymers and aluminosilicate shown in Fig. 10B indicates slight differences in chemical composition depending on the post-cure treatment of the samples. Importantly, no new peaks are observed when comparing the MISSE and Earth-control samples, only slight differences in relative peak heights. Fig. 10B focuses on the 2θ range of 20 to 37° to highlight the major crystalline peaks in the XRD patterns, and black squares on Fig. 9B represent peaks with clearly different intensity comparing the geopolymer samples to the LHS-1 aluminosilicate. XRD patterns from 5 to 70° (2θ) are available in the Supplementary Material. Significant differences in crystalline character at 26.7, 27.8, and 28.0 are present between the aluminosilicate LHS-1 powder and the geopolymer samples, likely indicating the dissolution of andesine and anorthite phases of the material (Javed et al., 2024). Comparing the geopolymer samples, the clearest visible difference is in the intensity of the andesine crystalline peak at 27.8 degrees (2θ). Similar to the metakaolin sample RMSE analysis, the RMSE statistical analysis indicates that the MISSE and Earth control patterns are much more similar than their

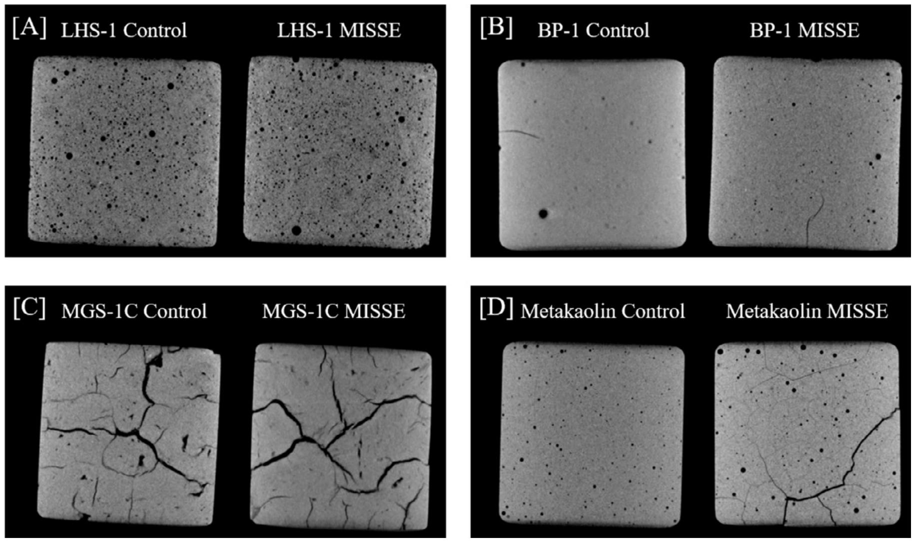

Fig. 6. X-ray tomography images of the internal structure of representative geopolymer samples. Each image represents the entirety of each selected sample. Comparisons of Earth Control and MISSE samples for (A) LHS-1, (B) BP-1, (C) MGS-1C, and (D) metakaolin geopolymers.

patterns in comparison to the aluminosilicate LHS-1 powder.

4. Discussion

4.1. Performance of extraterrestrial regolith simulant geopolymers

Geopolymer binders synthesized from lunar and Martian regolith simulants (LHS-1, BP-1, and MGS-1C) displayed excellent mechanical stability following six months of LEO exposure aboard the ISS. Distinct variations in mechanical performance are present based on individual geopolymer compositions. The chemical and mechanical property analyses determine that the statistically significant changes in the CS of the LHS-1 MISSE geopolymers are due to the pre-flight bakeout testing. Specifically, the measured CS increased for both the MISSE samples and the Temperature Swing samples compared to their control samples tested at the same time after synthesis indicates that the elevated temperature during bakeout testing is responsible. Increased sample exposure to elevated temperatures during both the pre-flight and simulated bakeout testing likely leads to additional curing, e.g. the formation of additional geopolymer reaction product (Mills et al., 2022). The XRD chemical analysis of the LHS-1 Earth Control, MISSE, and Temperature Swing samples show an overall reduction in crystalline peak intensity compared to the aluminosilicate precursor, indicating that dissolution has occurred. While outside the scope of this study, solid-state ^{29}Si and ^{27}Al NMR measurements could identify the formation of additional crosslinked geopolymer reaction product.

The measured increase in CS for the LHS-1, BP-1 and metakaolin Earth-control samples compared to the Immediate control samples indicates that the geopolymers continue to react at longer times. A one-year study of the mechanical properties of metakaolin geopolymers (Si/Al ratio of 1.8) initially cured at 60 °C reported an increase in compressive strength from 65 MPa at seven days to ~72.5 MPa at 365 days (Sun and Vollpracht, 2019). Compressive strength measurements of lunar regolith simulant geopolymers in the literature are often limited to a single time point (typically 7 or 28 days), however results in this study suggest that additional kinetic measurements of compressive strength are required to understand the mechanism for long-term gains in compressive strength.

4.2. Darkening of metakaolin geopolymers

Metakaolin geopolymer MISSE samples visually darkened during long-term LEO exposure and cracked during bakeout testing. Exposure to ultraviolet radiation can lead to oxide-darkening in the optical properties of amorphous silicon oxide materials, often through the growth and propagation of point defects originally present in the material (Li et al., 2017b). Metakaolin geopolymers intended for use as coatings also darkened when subjected to lab-based combined ultraviolet exposure and high-energy electron exposure (Cesul et al., 2014). Slight darkening is also observed for the MGS-1C samples which are enriched with smectite clay. Visible color change is a potential concern for the heat management of extraterrestrial structures, as changes in the IR reflectivity of the geopolymer may change the material behavior from a reflector to an absorber.

During the bakeout testing, cracking of metakaolin geopolymer samples was observed after the vacuum step when the samples are returning from low pressure and a temperature of 60 °C to ambient temperature and pressure. Importantly, a temperature jump from 60 to ~22 °C also occurred at the end of the curing protocol without any observed cracking. Therefore, the combined change in

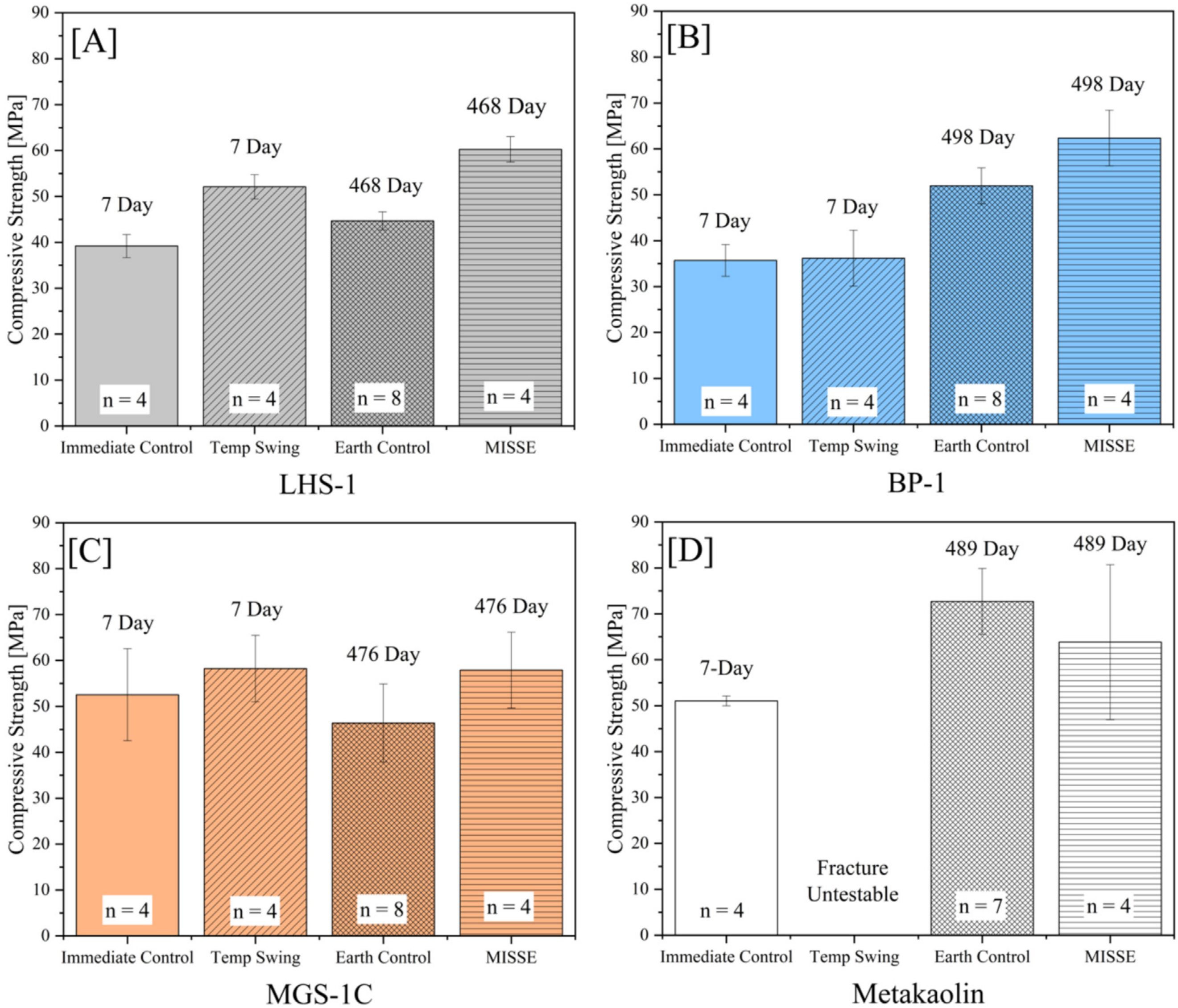

Fig. 7. Measured compressive strength of geopolymer samples following varied environmental testing including (A) LHS-1, (B) BP-1, (C) MGS-1C, and (D) metakaolin geopolymer samples. Error bars represent one standard deviation from the mean, and the sample size for each sample set (n) is included at the base of each bar The time after synthesis of compression testing is listed on the figure above each bar.

Table 5
Statistical two-tailed *t*-test results using two sample unequal variance for the comparison of MISSE and earth-control compressive strength. T-tests resulting in a p-value of 0.05 or less represent cases with a statistically significant difference in material properties (bold text).

Aluminosilicate	MISSE vs earth control	Immediate control vs temperature swing
LHS-1	**0.0003**	**0.0004**
BP-1	**0.032**	0.89
MGS-1C	0.063	0.40
Metakaolin	0.37	N/A

pressure and temperature is believed to initiate the cracking. The similarity of MISSE and Earth Control XRD patterns indicates that cracking is likely not initiated by the formation of a new reaction product or crystalline phase in the material during the bakeout testing. The identical bulk densities of samples subjected to different treatments (Table 4) suggest that cracking is not due to measured outgassing or volatile loss within the samples. The metakaolin geopolymers have a significantly lower solids mass fraction (40%) compared to the extraterrestrial regolith simulants (60–70%) to achieve the desired nominal Si/Al ratio of 2. Potential reasons for cracking of the metakaolin geopolymers include the expansion of water trapped in micropores of the binder leading to matrix cracking or differences in coefficients of thermal expansion between the undissolved metakaolin in the binder and the geopolymer reaction product phase.

4.3. VTVL pad construction

Vertical takeoff vertical landing (VTVL) pads will be a critical component of future extraterrestrial base camps.

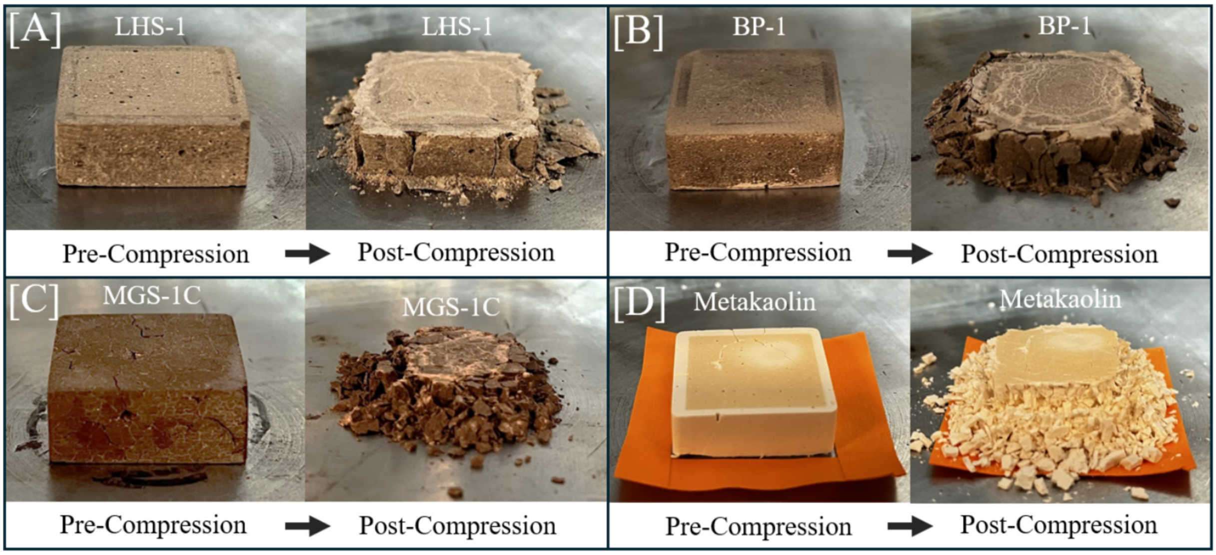

Fig. 8. Images of MISSE geopolymer samples immediately before and after compression testing for representative samples of (A) LHS-1, (B) BP-1, (C) MGS-1C, and (D) metakaolin geopolymers.

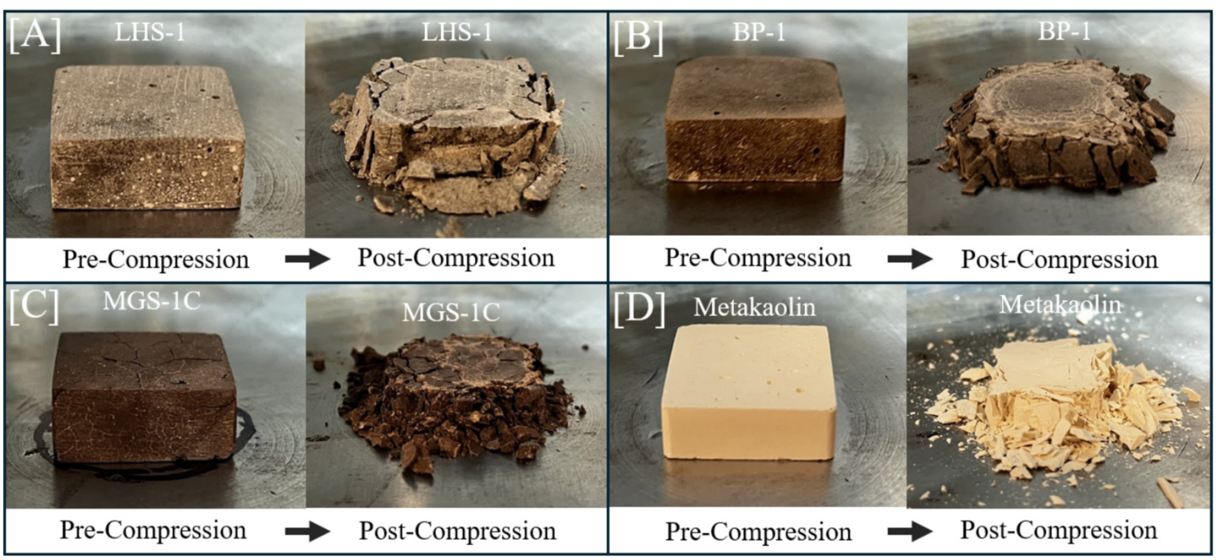

Fig. 9. Images of Earth Control geopolymer samples immediately before and after compression testing for representative samples of (A) LHS-1, (B) BP-1, (C) MGS-1C, and (D) metakaolin geopolymers.

This section provides an example calculation of the mass requirements to synthesize a bulk amount of LHS-1 geopolymer concrete for future VTVL construction. Recently, Jehn et al. modeled a circular VTVL pad of 40 m diameter and 1 m thickness, determining that the maximum stress experienced at the top surface of the pad from plume pressure is 365 kPa and the maximum stress from leg impacts and bearing is 322 kPa (Jehn and Dreyer, 2025). Stresses on the order of hundreds of kPa are well within the tolerable range of all geopolymer compositions in this study. Proposed VTVL pad thicknesses in the literature range from 12 mm thickness (Ferguson et al., 2018) to the 1 m thickness suggested by Jehn et al. (Jehn and Dreyer, 2025).

A lunar geopolymer concrete would be comprised of the lunar regolith, aggregate, and the alkaline activating solution comprised of water, sodium hydroxide, and fumed silica. Materials available for ISRU are assumed to be aggregate material, powdered lunar regolith, and water. Materials to be transported include sodium hydroxide and fumed silica. Table 7 provides material mass estimates to construct one m^3 of geopolymer concrete from the LHS-1 geopolymer in this study with a typical coarse aggregate content of 75% of the volume of the final material. An aggregate volume fraction of 75% is chosen as a reasonable upper limit based on typical terrestrial formulation of concrete formed from Ordinary Portland cement (Amparano et al., 2000) and commonly synthesized geopolymer con-

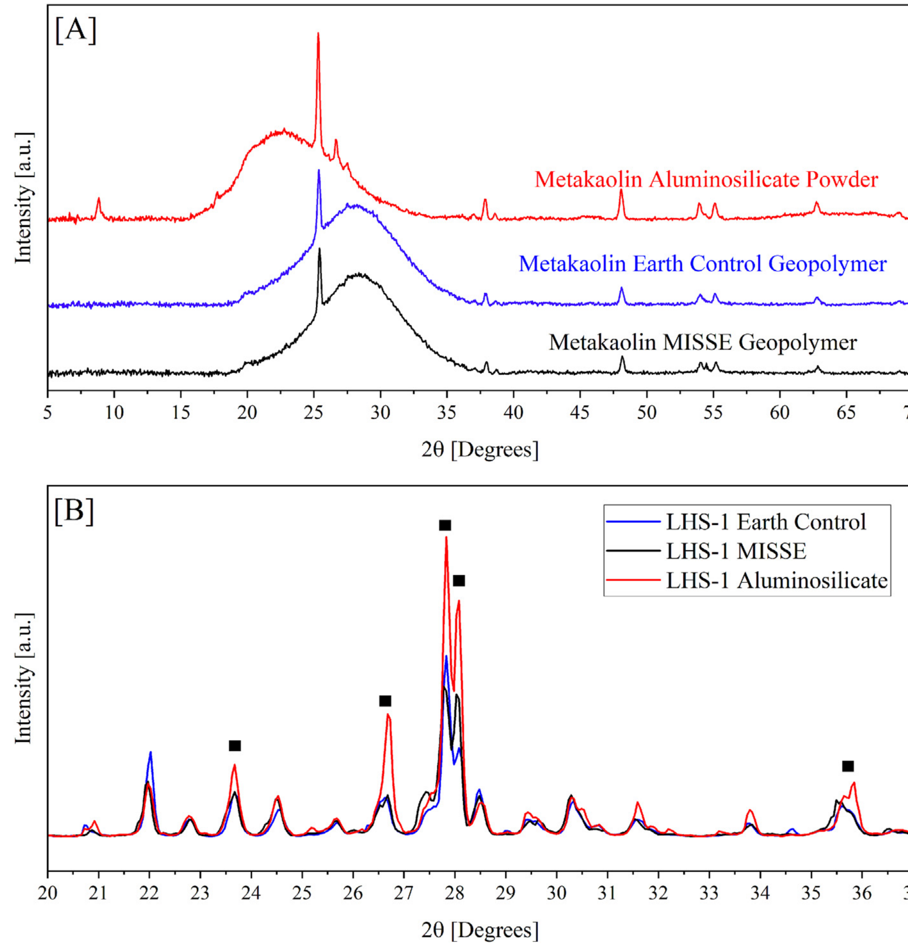

Fig. 10. XRD patterns of (A) metakaolin samples and (B) LHS-1 samples including (blue) the earth-control geopolymer, (black) the MISSE geopolymer, and (red) the raw aluminosilicate powder. The MISSE and Earth Control samples selected for XRD examination are synthesized in the same cube mold. Black squares in (B) denote peaks of interest in the results and discussion.

Table 6
Statistical root mean square error (RMSE) analysis of XRD patterns of metakaolin and LHS-1 geopolymer samples.

Metakaolin	
Comparison	Root Mean Square Error (RMSE)
MISSE vs Earth Control	0.37
MISSE vs Aluminosilicate	2.76
Earth Control vs Aluminosilicate	2.92

LHS-1	
Comparison	RMSE
MISSE vs Earth Control	6.11
MISSE vs Aluminosilicate	13.18
Earth Control vs Aluminosilicate	14.14

cretes (Nuaklong et al., 2016; Joseph and Mathew, 2012). The density of basalt is used as a surrogate for the density of lunar aggregate (2630 kg/m^3). As mentioned in the introduction, water required for geopolymerization is up to 98% recyclable as it is not chemically sequestered in the reaction product (Wang et al., 2017). The volume of geopolymer concrete required to create a 40 m diameter VTVL pad with thickness of 1.2 cm is 15.0 m^3. The calculated required lift mass of 60.7 kg per 1 m^3 of lunar geopolymer indicates the potential feasibility of this proposed construction material.

4.4. Future work

The combined results for all extraterrestrial simulant geopolymers (LHS-1, BP-1, and MGS-1C) indicate that geopolymer binders maintain excellent mechanical stability following six months of LEO exposure aboard the ISS, chosen for this study as an intermediate stage towards future testing on the lunar surface. This key result motivates the continued research and development of extraterrestrial regolith simulant geopolymers with optimized characteristics for lunar surface applications including but not limited to (1) reduced payload requirements from Earth by minimizing activator content, (2) resistance to degradation under micrometeorite impact, and (3) stability under the more intense temperature swings present on the lunar sur-

Table 7
ISRU and lift mass requirements to synthesize 1 m^3 of LHS-1 geopolymer concrete with an aggregate volume fraction of 0.75.

	ISRU mass requirements			Lift mass requirements	
Geopolymer Concrete Volume	LHS-1 Mass [kg]	Aggregate Mass [kg]	Water Mass [kg]	NaOH Mass [kg]	SiO_2 Mass [kg]
1 m^3	353	1970	90.9	25.3	35.4

face reaching temperatures as low as −223 °C. The geopolymers in this study are formulated to produce cured samples with high compressive strength, not specifically to minimize payload requirements. Lunar regolith simulant geopolymers can be synthesized with up to 83 wt% solids, as compared to the 70 wt% solids used in the LHS-1 geopolymer in this study (Pilehvar et al., 2020). Additionally, lunar base materials must remain functional for years following construction, significantly longer than the six-month period of study in this work. When future lunar surface durability testing studies are planned, it is recommended to measure performance over several years to assess the impact of long-term radiation and temperature fluctuations. Alternatively, accelerated aging experiments can be designed to target the effects of temperature fluctuations.

Recognizing that the in-situ resources available as well as the environments differ between the lunar and Martian surface, and that the environments are different to LEO conditions suggests further research is needed for each specific situation. For the lunar surface, temperatures can range from −233 °C within the shadowed polar craters at night to 123 °C at the equator during daytime, compared to the smaller temperature range experienced in the study (−11.75 °C to 35.0 °C) (Vaniman et al., 1991). Further, due to the Moon's relatively small magnetic field and lack of atmosphere, high-energy protons from solar flares, alpha particles, and heavier nuclei originating from high-energy galactic cosmic rays directly bombard the lunar surface, all of which can damage materials (Sherwood and Toups, 1992). Nonetheless, a clear point of similarity between the LEO environment studied in this work and the lunar surface is the exposure to vacuum conditions. Micrometeorites also bombard the lunar surface continually, and any future lunar construction materials must be resistant to this method of degradation that is not present in the LEO environment on the timescale of the measurements performed here. Meanwhile, the Martian surface ranges in temperature from −98 °C overnight to −8 °C determined via at the Mars Exploration Rover Opportunity at the Meridiani Planum site (Spanovich et al., 2006). In contrast to LEO and the lunar surface, the Martian atmosphere is almost exclusively composed of CO_2 and has a significantly reduced density compared to the Earth's atmosphere (610 Pa on Mars vs 101 kPa on Earth) (Petrosyan et al., 2011; Collins et al., 2023). While the current study probes the LEO effects of long-term vacuum exposure, constant temperature swings, and significant UV radiation exposure, future geopolymer materials testing targeting lunar or Martian use cases must take into account these differences in environmental conditions.

The further development of processing methods to cure lunar geopolymer binders on the lunar surface with high strength remains a critical challenge. Significant limitations to compressive strength gain are observed under vacuum conditions as compared to control samples in earth laboratory conditions and water recovery during curing is necessary (Mills et al., 2022; Collins et al., 2022a; Davis et al., 2017). Additionally, the reactions responsible for geopolymer strength development are temperature sensitive, and slow considerably at low temperature (Rovnaník, 2010). However, advances in the curing of lunar regolith simulant geopolymers via microwave irradiation offers a potential solution, and such approaches are the subject of future research (Sun et al., 2021, Shi et al., 2020).

5. Conclusions

Measurements of geopolymer durability in low-earth orbit for a period of six months are completed as part of NASA's MISSE-20 experiment on the International Space Station. High-strength geopolymer samples formed from the LHS-1 and BP-1 lunar regolith simulants and the MGS-1C Martian simulant resist degradation and maintain mechanical strength despite exposure to vacuum, constant temperature oscillations of 15 °C per Earth orbit with an absolute temperature range of −11.75 °C to 35.0 °C, and high doses of radiation (cumulative UV fluence of 15.0 kJ cm^{-2}). Statistically significant differences observed in mechanical properties between control samples and samples exposed to LEO originate in the high-temperature pre-flight bakeout testing, resulting in increased compressive strength for LHS-1 derived MISSE samples (MISSE Samples: 60.3 MPa vs. Earth-control samples: 44.7 MPa, ~35% increase). BP-1 and MGS-1C geopolymers also displayed an average increase in compressive strength when compared to their Earth-control counterparts (20% and 25% respectively). Considerable degradation of the metakaolin geopolymers is observed in both macroscale cracking of the samples and darkening of the exposed surface because of oxide attack. Cracking is confirmed to occur because of pre-flight temperature swing and vacuum testing, not because of LEO exposure. In conclusion, lunar and Martian regolith simulant geopolymers are shown to maintain high compressive strength and resist degradation following six-months of LEO exposure, confirming the aptitude of this class of materials for space applications. There is a timely need to extend this research and develop processing protocols and testing conducted on lunar regolith and under environmental conditions more similar to the lunar south pole, where more pronounced temperature swings and micrometeorite impacts can potentially affect geopolymer durability.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.asr.2026.02.080.

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